Facial Expression Analysis

Decode emotional expressions as they happen.

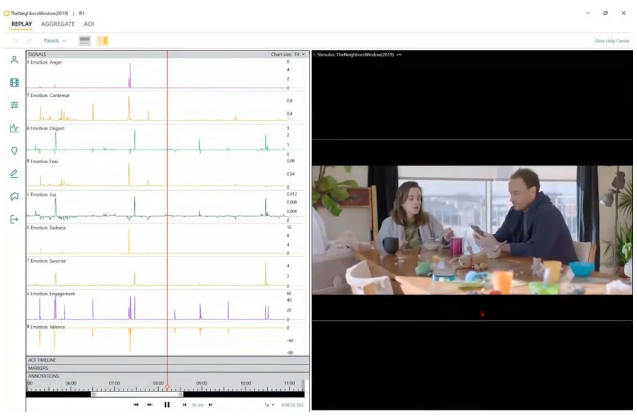
Works with

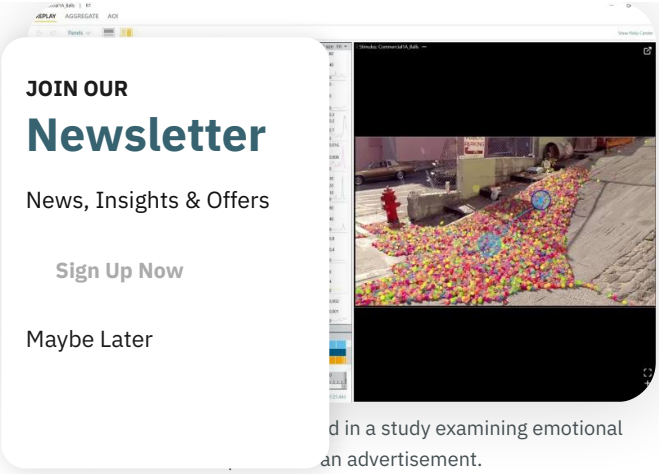
iMotions Lab **Remote Data Collection**

AI Powered Facial Expression Analysis

Our face displays our outward emotional expressions – giving a view of how we show our inner emotional state. These expressed emotional states are detected in real time using fully automated computer algorithms that record facial expressions via webcam.

Tracking facial expressions can, when used in controlled contexts and in collaboration with other biosensors, be a powerful indicator of emotional experiences. While no single sensor is able to read minds, the synthesis of multiple data streams combined with strong empirical methods can begin to reach in that direction.





d in a study examining emotional
an advertisement.

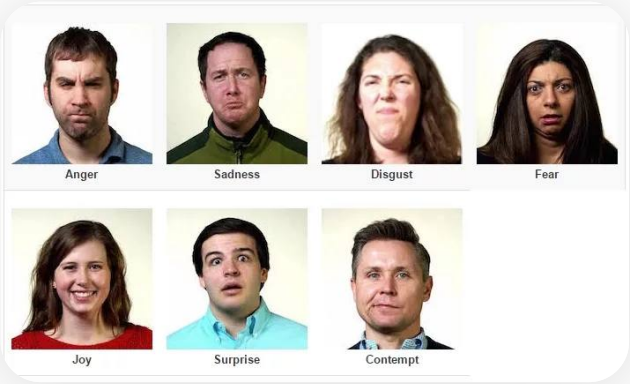
Live or post **automatic facial coding** from any video

The iMotions Facial Expression Analysis Module seamlessly integrates the leading automated facial coding AI Affectiva's AFFDEX. Using a webcam, you can live synchronize expressed facial emotions with stimuli directly in the iMotions software. If you have recorded facial videos, you can simply import videos and carry out the analysis. Gain insights via built-in analysis and visualization tools, or export data for additional analyses.

Detect 7 core emotions

iMotions' facial expression analysis module provides real-time insights by tracking subtle facial muscle movements to identify seven core emotions: **Joy**, **anger**, **fear**, **surprise**, **sadness**, **contempt**, **and disgust**. The methodology is built on the **Facial Action Coding System (FACS)**, the objective framework developed by Dr. Paul Ekman and Wallace Friesen.

Researchers can leverage this data for applications such as **user experience testing, market research, and mental health assessmen**t, enhancing product design, marketing strategies, and overall well-being.



The 7 Core Emotions: Joy, Anger, Fear, Surprise, Sadness, Contempt and Disgust



Valence and Engagement are crucial metrics for quantifying an experience.

Get Valence and Engagement Statistics For Your Content

Valence and engagement are crucial metrics for understanding emotional responses. **Valence** represents the overall emotional tone, ranging from negative to positive. It helps researchers assess user experiences, product preferences, and brand perception. **High valence indicates positive emotions**, while **low valence suggests negative feelings.**

On the other hand, **engagement** measures the level of expressiveness and involvement. It reflects how actively an individual responds to stimuli, such as advertisements, videos, or interactive content. By tracking valence and engagement, businesses can optimize marketing strategies, improve user satisfaction, to name a few.

Track the Movement of the Face

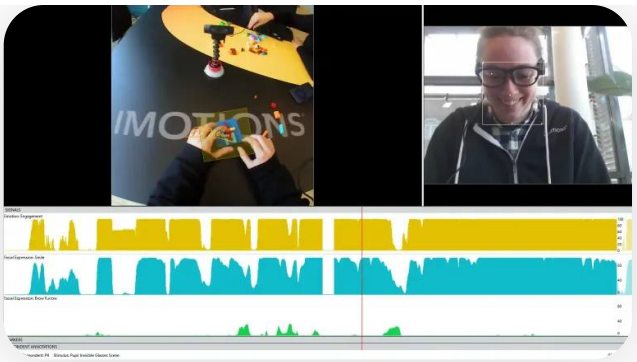
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A study investigating emotional expressions while building LEGO.

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Facial Expression Analysis
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Get a Demo

We'd love to learn more about you! Talk to a specialist about your research and business needs and get a live demo of the capabilities of the iMotions Research Platform.

Request Demo

FAQ

Here you can find some of the questions we are asked on a regular basis. If you have questions you cannot find here, or elsewhere on our website, please contact us here.

What is Facial Expression Analysis (FEA)?

What is the Facial Action Coding System (FACS)?

How does Affectiva use FACS?

Can I buy the AFFDEX SDK through iMotions?

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EX or Realeyes algorithms

e expressions

h AFFDEX and 29 expressions with Realeyes

Access data about overall expressiveness – whether the participant is neutral or engaged

- **Built-in QA**

  Integrated quality assurance tools

- **Post-import video**

  Import externally recorded videos for post processing of facial expressions

- **Head Orientation**

  Access head rotation data, such as yaw (up/down), pitch (left/right), and roll

- **Interocular Distance**

  Distance between two outer eye corners for estimation of distance from screen

Latest publications using iMotions FEA (Facial Expression Analysis) module

Cognitive Behavioral Therapy Reduces Symptom Severity and Normalizes Neurophysiological and Attentional Reactivity in Anorexia Nervosa: A Randomized Controlled Trial

Open Access Peer-Reviewed 13/03/2026 **Beykoz University + 3**

Confronted with mathematics: the impact of pre-service elementary school teachers' self-concept on emotional and physiological responses to mathematics exposure

Open Access Peer-Reviewed 10/03/2026 **Ludwig-Maximilians-Universität München + 3**

Poker face and steady voice: Gender and reactions to emotional neutrality in crowdfunding

Open Access Peer-Reviewed 24/02/2026 **University of St. Gallen + 3**

Even Better than the Real Thing: How Imperfection Shapes Trust and Engagement with Digital Humans

Open Access Peer-Reviewed 06/01/2026 **University of Sydney + 3**

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Resources

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Facial Expression Analysis

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iMotions Remote Data Collection Guide

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Brochures



Affectiva for Facial Expression Analysis



RealEyes for Facial Expression Analysis



Student Guide to Data Analysis in iMotions Online

Science Resources

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AFFDEX 2.0: A Real-


Affectiva SDK: A Cross-Platform RealTime Multi-Face Expression Recognition Toolkit


Affectiva-MIT Facial Expression Dataset (AM-FED): Naturalistic and Spontaneous Facial Expressions Collected “In-the-Wild”


Automatic Detection of Sentimentality from Facial Expressions


Emotion Detection in Commercial Applications


Multimodal biometric data collection of interpersonal communication


Predicting Ad Liking and Purchase Intent: Large-scale Analysis of Facial Responses to Ads


Predicting Online Media Effectiveness Based on Smile Responses Gathered Over the Internet


Which CNNs and Training Settings to Choose for Action Unit Detection? A Study Based on a Large-Scale Dataset

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Eye Tracking
Screen Based

Eye Tracking VR
Virtual and mixed reality eye tracking

Eye Tracking Glasses
Eye tracking using glasses

Respiration
Measure breathing

EDA / GSR
Electrodermal Activity

EMG
Electromyography

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fNIRS
Measures brain activity through blood and oxygen changes

ECG
Electrocardiogram

GPS
Track location, movement and speed

Voice Analysis
Emotion and Voice Characteristics

Facial Expression Analysis
Facial Expression Analysis

FEA Multiface
FEA addon for multiface tracking

Remote Data Collection
Collect webcam eye tracking, FEA, Respiration and Voice

Webcam Eye Tracking
Webcam Eye Tracking

Webcam Respiration
Measure breathing via video

Inspiring Research



Multiface Analysis in Action: Advanced Methods for Studying Facial Expressions in Group Settings

ACADEMIA
8 min read

Dr. Divya Seernani
Updated 13/03/2026



Desire Before Delight: Why Wanting Drives Consumer Choice More Than Liking

CONSUMER INSIGHTS
10 min read

Vrushali Panjwani
14/01/2026



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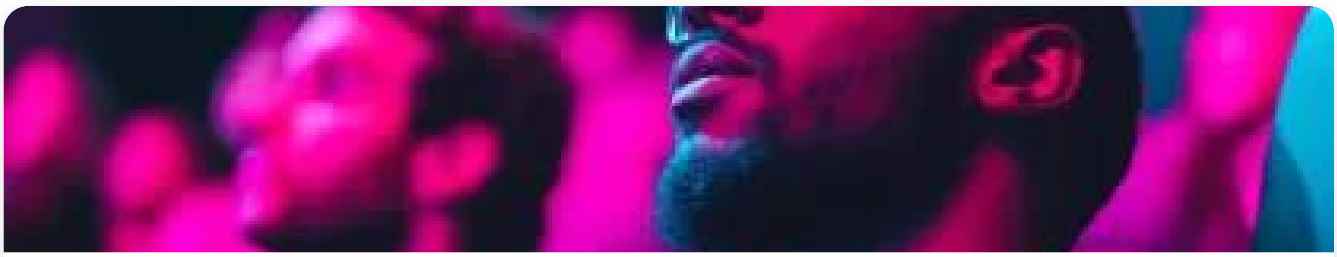
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Bryn Farnsworth

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